

Outline_COT / MEOW-style Outline Evaluation Metrics, Power, and Minimum Detectable Effect Report

現行 metrics 的數學盤點、paired comparison 樣本數推導、 $\Delta_{\min}$ 與 $N_{\min}$ 曲線

檔名：metric_power_curve_report.pdf

主要語言：繁體中文，technical terms 保留原文

主要設定：paired paper-level comparison, two-sided $\alpha=0.05$, power=0.8；附 power=0.9 / one-sided sensitivity

報告日期：2026-05-20

source-confirmed 本報告以 Sologa/Outline_COT 的 AGENTS.md、docs/meow_evaluation_assets.md、scripts、prompts 與 results summary JSON 為主要 source artifacts。未在現行 source 中確認的 metric family 以 **theoretical-assumption** 或 **source-pending** 標註。

1. Executive summary

source-confirmed 現行 Outline_COT / MEOW-style evaluation 不是單一 continuous score。source inspection 確認目前主要可操作 metrics 至少包含三條路徑：一、programmatic Structural Distance；二、prompt-driven upstream repo-defined 6D LLM-as-a-Judge；三、auxiliary Reference Reward / $F\beta$ -like ref-overlap composite。Paper-level 5-criterion judge 與 upstream 6D judge 不應被混為同一個 metric。

theoretical-assumption 主要 inferential unit 應是 paper-level N 。section、tree node、reference pair、judge dimension 都是 nested / clustered observations，不應直接當成獨立樣本數。把 6 個 judge dimensions 當成 6N 會低估標準誤，除非明確建模 within-paper / within-outline correlation。

theoretical-assumption 在 two-sided $\alpha=0.05$ 、power=0.8 的 paired continuous approximation 下，MDE multiplier 是 $2.802 / \sqrt{N}$ 。因此 $N=4$ 只能檢出約 1.401 σd 的差距； $N=100$ 可檢出約 0.280 σd ； $N=200$ 約 0.198 σd 。這是最低可檢出差異，不是保證效果存在。

empirical-estimated 兩個已存在的 3-paper summary runs (gpt-5.4-xhigh-3papers 與 gemini-3.1-pro-preview-3papers) 可提供非常弱的 paired σd calibration：normalized Structural Distance improvement 的 paired σd 約 0.054；6D judge composite 的 paired σd 約 0.411。因為只有 $N=3$ ，這些數值只可作為 sanity check，不可作為最終 power planning 的唯一依據。

theoretical-assumption $N=4$ 應視為 smoke / exploratory signal。即使 p-value 顯著，也容易受單一 paper、judge noise、gold-outline mismatch、prompt formatting、API stochasticity 影響；不宜寫成穩健的 model superiority claim。

theoretical-assumption 實務建議是：若想比較兩個模型 / prompt / run 的主要效果，先 preregister 一個 primary endpoint (例如 normalized Structural Distance 或 6D composite mean)，再把其他 dimensions 當 secondary endpoints 並使用 Holm 或 BH 控制 multiple comparisons。若預期 judge-score paired σd 約 1.0， $N=50$ 約能檢出 0.40 分； $N=100$ 約 0.28 分； $N=25$ 只能檢出約 0.56 分以上。

2. What “statistically significant effect” means here

theoretical-assumption 本報告中的 statistically significant effect 指的是：在事先指定的 test、 α 、effect direction 與 sample size 下，觀察到的 test statistic 足以拒絕 null hypothesis。這不是效果為真或未來會重現的保證。Power 是在假設真實差異 Δ 存在時，檢出顯著結果的機率；MDE 是在給定 N 、 α 、power 與 σd 下，大致能穩定檢出的最小差距。

theoretical-assumption 對 paired comparison，令每篇 paper 的差異為 $d_i = \text{score}_i(A) - \text{score}_i(B)$ 。對 lower-is-better distance，建議先轉成 improvement，例如 $d_i = \text{distance}_i(B) - \text{distance}_i(A)$ ，使正值代表 A 較好。若直接用原始 distance，差異方向需要反過來解讀。

Paired continuous MDE:

$$\Delta_{\min}(N) = (z_{1-\alpha/2} + z_{\text{power}}) \cdot \sigma_d / \sqrt{N}$$

$$N_{\min}(\Delta) = [(z_{1-\alpha/2} + z_{\text{power}}) \cdot \sigma_d / \Delta]^2$$

theoretical-assumption two-sided $\alpha=0.05$, power=0.8 時， $z_{0.975} + z_{0.8} = 1.960 + 0.842 = 2.802$ 。power=0.9 時，常數變為 3.242。因此把 power 從 0.8 提到 0.9 會讓 $\Delta_{\min}$ 增加約 15.5%，或讓所需 N 約增加 33.8%。

Independent two-sample equal-n contrast:

$$\Delta_{\min}(n \text{ per arm}) = (z_{1-\alpha/2} + z_{\text{power}}) \cdot \sqrt{(\sigma_A^2/n + \sigma_B^2/n)}$$

若 $\sigma_A = \sigma_B = \sigma$ ，則 $\Delta_{\min} = (z_{1-\alpha/2} + z_{\text{power}}) \cdot \sqrt{2} \cdot \sigma / \sqrt{n}$ 。

theoretical-assumption paired design 的核心優勢是消除 paper difficulty 的共同變異；其效果大小取決於 paired difference 的 σ_d ，而不是單一 condition score 的 σ 。若兩條件在同一批 papers 上高度相關， σ_d 會明顯小於 independent comparison 的標準誤。

3. Source verification and metric inventory

source-confirmed 下表列出本報告實際檢查的主要 source artifacts。Source paths 以 repo 內路徑呈現；SHA 是 GitHub connector 回傳的 file sha 或可追蹤版本資訊。

source artifact	revision / timestamp	role in report
AGENTS.md	Sologa/Outline_COT main, sha ba06cbc09d504e82789e168eb88140dbbb21eaf4	Repo purpose, refs/results contract, blind generation/evaluation contract, MEOW evaluation asset rules.
docs/meow_evaluation_assets.md	sha f57689bf073f6029d2cfa5ab147b543c921c39b9	Metric status: upstream 6D judge, Structural Distance, auxiliary Reference Reward, paper 5D vs upstream 6D distinction.
scripts/combine_scores.py	sha 66c71a69f84a1aa936e8c4418befca031f16bf43	Normalized tree-shape distance and call to article_reward.
scripts/evaluate_chatgpt_meow_blind_batch.py	sha 4672473e969360c491a9ba50348ee05eefb4a97a	Blind eval resolution, SCORE_KEYS, structural debug, judge outputs, summary schema.
prompts/meow_llm_judge_6d_source_system/user.txt	sha f2602c... and d6274b...	6D judge prompt; 0-10 with 0.5 increments.
scripts/ref_reward.py	sha 576a0566cbbcb44be852e79e37c4ab847	F_beta, length penalty, section matching by numbering, article-level reward.
scripts/evaluate_pair_rewards.py	sha 45f14fa25bc2405c1f41b9e332e3362adc32fbd6	Batch output for shape_distance and reward.

source artifact	revision / timestamp	role in report
scripts/compare_outlines_shape.py	sha 68b337a1dd180065784a18208a2e3dcb845acd70	Raw shape-only TED path; labels omitted.
upstream cedricshan/Survey-Outline-Evaluation-Benchmark/ scripts/evaluate_llm.py	sha b1491134062c3a481a3dc167c9d8af29fb19a974	Upstream 6D judge implementation mirrored locally.
results/_summaries/gpt-5.4-xhigh-3papers and gemini-3.1-pro- preview-3papers	generated_at 2026-04-08, judge_model gpt-5-nano	Representative per-paper empirical paired data used only as weak calibration.

source-confirmed AGENTS.md 與 docs/meow_evaluation_assets.md 明確要求區分 refs 與 results、blind-outline evaluation 的 results-first contract、upstream vendor mirror、local promoted evaluation entrypoints、以及 paper 5D judge 和 upstream 6D judge 的差異。這對 power analysis 很重要，因為不同 metric 的 unit、range、direction 和 stochastic noise 不同。

source-pending 使用者列出的 scripts/cli_judge_backends.py 在目前 Sologa/Outline_COT main 上未找到；第三方 repo 路徑在 source docs 中是 resources/Survey-Outline-Evaluation-Benchmark/，不是 third_party/repos/。本報告對未找到的檔案不作 source-confirmed claim。

Metric inventory table

表 1 是 inferential planning 的主表。為了避免把 source-confirmed 與 theory-only 混淆，每列都含 status 欄位。

metric name	source file / provenance	value range	higher/lower	metric family / type	per-paper observable	primary test	robustness check	paired valid	empirical variance estimable	status
Normalized Structural Distance / shape_distance	scripts/combine_scores.py; scripts/evaluate_chatgpt_meow_blind_batch.py ; docs/meow_evaluation_assets.md	0 to approximately 1 in normalized path: edit_ops / max(node counts); non-negative	lower is better	bounded or quasi-bounded continuous distance; nonparametric paired outcome	one normalized tree edit distance between model outline and reference outline, using only section levels/tree shape	paired t-test on distance reduction if N moderate; otherwise paired permutation test	Wilcoxon signed-rank, sign test, paired bootstrap CI; inspect outliers by paper	yes, when same paper/reference is evaluated under both conditions	yes from per-paper eval JSON; current report includes N=3 weak paired estimate from two runs	source-confirmed
Raw shape-only Zhang-Shasha TED	scripts/compare_outlines_shape.py	non-negative integer or float edit count, unnormalized	lower is better	count distance / tree edit count	raw shape edit distance; labels ignored/all nodes identical	paired permutation or Wilcoxon; count model if needed	normalize by node count, paired bootstrap, sensitivity by outline length	yes	yes if raw compare outputs exist; not the main blind-eval normalized path	source-confirmed but auxiliary/ inconsistent scale
LLM judge: 结构_信息快速定位	prompts/meow_llm_judge_6d_source_user.txt; scripts/evaluate_llm.py; scripts/evaluate_chatgpt_meow_blind_batch.py	0-10 with possible 0.5 increments	higher is better	bounded ordinal score, often analyzed with interval approximation	one judge score per generated outline per paper	paired t-test under interval approximation	paired permutation, Wilcoxon, ordinal cumulative-link mixed model; repeat judge calls if possible	yes, if same judge protocol scores both conditions on same paper set	yes from eval JSON; judge stochastic variance needs repeated judging	source-confirmed

metric name	source file / provenance	value range	higher/lower	metric family / type	per-paper observable	primary test	robustness check	paired valid	empirical variance estimable	status
LLM judge: 结构_详略得当	same 6D judge prompt/code	0-10 with possible 0.5 increments	higher is better	bounded ordinal score	per-paper dimension score	paired t-test or permutation on paired score differences	Wilcoxon / ordinal model / bootstrap	yes	yes	source-confirmed
LLM judge: 内容_章节互斥性	same 6D judge prompt/code	0-10 with possible 0.5 increments	higher is better	bounded ordinal score	per-paper dimension score	paired t-test or permutation	Wilcoxon / ordinal model / bootstrap	yes	yes	source-confirmed
LLM judge: 内容_逻辑深度	same 6D judge prompt/code	0-10 with possible 0.5 increments	higher is better	bounded ordinal score	per-paper dimension score	paired t-test or permutation	Wilcoxon / ordinal model / bootstrap	yes	yes	source-confirmed
LLM judge: 内容_学术价值	same 6D judge prompt/code; docs/meow_evaluation_assets.md identifies this as upstream extra dimension vs paper 5D	0-10 with possible 0.5 increments	higher is better	bounded ordinal score	per-paper dimension score	paired t-test or permutation	Wilcoxon / ordinal model / bootstrap	yes	yes	source-confirmed as upstream 6D, not paper 5D
LLM judge: 语用_描述性与简洁性	same 6D judge prompt/code	0-10 with possible 0.5 increments	higher is better	bounded ordinal score	per-paper dimension score	paired t-test or permutation	Wilcoxon / ordinal model / bootstrap	yes	yes	source-confirmed
LLM judge composite mean / judge_average_scores aggregation	scripts/evaluate_chatgpt_meow_blind_batch.py; scripts/evaluate_llm.py summaries	0-10 if averaging the six 0-10 dimensions	higher is better	aggregated composite of correlated ordinal dimensions	recommended: average six scores within each paper, then compare papers; summary averages alone are insufficient for paired tests	paired t-test or permutation on per-paper composite differences	multivariate paired test, dimension-specific Holm/BH, bootstrap preserving within-paper dimension vector	yes, but dimensions are not independent samples	yes if per-paper judge_scores are present	source-confirmed
Reference Reward / R_article	scripts/ref_reward.py; scripts/combine_scores.py; scripts/evaluate_pair_rewards.py	floored at -1; upper approximately 1; composite of F_beta and length penalties	higher is better	bounded ratio/ count-derived composite; F_beta over reference-set overlap plus penalties	article-level reward after matching sections mainly by numbering; averages over true sections with refs	paired permutation or paired bootstrap on per-paper R_article differences	hierarchical bootstrap over papers and sections; simulation over TP/FP/FN/ref-length scenarios	yes if same papers and same reference outlines	yes if pair reward JSONL exists; current dataset packaging gap requires explicit human outline JSONL	source-confirmed auxiliary metric
Section-level F_beta(refs)	scripts/ref_reward.py helper f_beta	0-1 before penalty	higher is better	F1/F_beta-like ratio metric from TP, predicted refs, true refs	not directly primary; section-level matched ref overlap; must aggregate by paper	paper-level bootstrap/ permutation after aggregation, not section-as-independent t-test	delta method or simulation over confusion-matrix/ ref-count scenarios; hierarchical bootstrap	yes only after paper-level clustering is respected	details available when ref_reward return_details is stored	source-confirmed helper, not current main table metric
Judge parse success / evaluation status	scripts/evaluate_llm.py; scripts/evaluate_chatgpt_meow_blind_batch.py	binary or categorical counts: success, partial_failure, failure; success_rate 0-100%	higher success is operationally better; not quality score	binary/proportion/ categorical operational metric	success indicator for each paper-condition evaluation	McNemar for paired success/failure; exact binomial or two-proportion if independent	paired bootstrap; inspect failure mechanisms	yes when same papers are attempted in both conditions	yes from status_counts/per-paper statuses	source-confirmed operational metric

metric name	source file / provenance	value range	higher/lower	metric family / type	per-paper observable	primary test	robustness check	paired valid	empirical variance estimable	status
Timing: structural_seconds / judge_seconds / total_seconds	scripts/evaluate_chatgpt_meow_blind_batch.py output fields	positive continuous seconds	lower is better for cost/latency, not quality	skewed continuous operational metric	per-paper timing fields	paired Wilcoxon or paired t on log seconds	bootstrap and median difference; stratify API model/backend	partially; timing depends on backend/load, not only paper	yes from eval JSON but environmental confounding high	source-confirmed operational metric
Pearson / Spearman / Kappa / Fleiss Kappa	not found in inspected current metric files	metric-specific	metric-specific	correlation/agreement	none in current code path	Fisher z or asymptotic/simulation if later added	bootstrap over papers; rater-level model	depends on design	not from current inspected run outputs	theory-only / no source-confirmed current metric
scripts/cli_judge_backends.py backend metrics	requested file not found on current Sologa/Outline_COT main; repo search returned no match	unknown	unknown	pending	pending	pending source verification	pending source verification	pending	pending	source-pending

4. Mathematical theory by metric family

4.1 Structural Distance / tree-shape distance

source-confirmed normalized Structural Distance 的主路徑在 combine_scores.py 中把 outline section list 依 level 建成 zss.Node tree，所有非 root 節點標籤都設為相同的 n，然後以 Zhang-Shasha tree edit distance 的 raw edit operations 除以兩棵樹最大 node count。這表示它只看 tree shape / level nesting，不看 title wording、numbering text、references、topic semantics 或 citation correctness。

theoretical-assumption 統計上可把 normalized shape_distance 視為 bounded continuous / semi-continuous paired distance。主要比較時建議定義 improvement $d_i = \text{distance}_{B,i} - \text{distance}_{A,i}$ ，正值代表 A 更接近 reference。若 $N \geq \text{約 } 25$ 且 outliers 不嚴重，可用 paired t-test 搭配 95% CI；N 小或 distribution skewed 時，用 paired permutation test、Wilcoxon signed-rank 或 sign test 做 robustness。

4.2 LLM-as-a-Judge ordinal / bounded score metrics

source-confirmed upstream repo-defined judge 是 6D prompt-driven metric，每個 dimension 0-10 且可含 0.5 分。這是 bounded ordinal score，不應被不加說明地當成普通 Gaussian score。實務上，若研究問題是平均分數差、且評分刻度有 21 個可能值，paired t-test 的 interval approximation 可作為 primary analysis，但應搭配 permutation / Wilcoxon / ordinal model。

theoretical-assumption 若保守處理 ordinal 性質，可用 cumulative-link mixed model： $\text{score_category} \sim \text{condition} + (1 | \text{paper})$ ，必要時加 judge-run random effect。若同一 outline 會被同一 judge model 重複評多次，應分解 paper variance、condition effect、judge stochastic variance；否則 single-call judge score 的 σ_d 會把模型差異與 judge noise 混在一起。

4.3 Bounded continuous metrics

source-confirmed normalized Structural Distance 和 Reference Reward 都屬於 bounded / quasi-bounded outputs。若分數接近邊界，例如 distance 接近 0 或 reward 接近 1，raw-scale normal approximation 可能失真。可用 logit transform 於 (0,1) metric、arcsine sqrt transform 於 proportions，或直接用 paired bootstrap / permutation 在 raw scale 報告 CI。

4.4 Proportion / binary / count-derived metrics

source-confirmed current scripts 輸出 success/failure/status_counts 與 success_rate。若比較兩條件在同一批 papers 上是否成功解析 / 是否 pass 某 threshold，應使用 McNemar-style paired analysis，而不是 independent two-proportion test。只有當兩組 papers 完全不同或設計無法配對時，才使用 independent two-proportion formula。

Paired binary planning approximation:

令 $q = p_{10} + p_{01}$ 為 discordant-pair fraction， $\delta = p_{10} - p_{01}$ 。

$\delta_{\min} \approx (z_{1-\alpha/2} + z_{\text{power}}) \cdot \sqrt{q} / \sqrt{N}$ 。

實務上更建議直接 simulation: 對 $(p_{11}, p_{10}, p_{01}, p_{00})$ 做 multinomial sampling，套用 exact/asymptotic McNemar。

Independent two-proportion planning formula:

$n \text{ per arm} \approx [z_{1-\alpha/2} \sqrt{(2 \bar{p}(1-\bar{p}))} + z_{\text{power}} \sqrt{(p_1(1-p_1)+p_2(1-p_2))}]^2 / (p_1-p_2)^2$ ，其中 $\bar{p}=(p_1+p_2)/2$ 。

4.5 Precision / Recall / F1 / ratio metrics

source-confirmed Reference Reward 使用 F_β over reference sets， $\beta=0.5$ ，並加入 length penalty、missed-section penalty、spurious-section penalty。 F_β 是 TP、pred_count、true_count 的非線性 ratio，因此不能天真把每個 section F_β 當獨立 normal score；section nested in paper，且 section matching 由 numbering 驅動。

theoretical-assumption 建議兩種估計曲線的方法：一、hierarchical bootstrap，先抽 paper，再在 paper 內保留 section/ref 結構，重新計算 article_reward 或 aggregate F_β ；二、scenario simulation，指定每篇 paper 的 true ref counts、pred ref counts、TP overlap、miss/spurious rates，對 confusion matrix 變動做 Monte Carlo，得到 paired difference distribution 與 power curve。

4.6 Correlation / agreement metrics

theoretical-assumption 目前 inspected source 沒有 source-confirmed Pearson、Spearman、Kappa 或 Fleiss Kappa。若未來加入 correlation difference，可用 Fisher z: $z_r = \text{arctanh}(r)$ ， $SE \approx 1/\sqrt{N-3}$ ；兩個 independent correlations 的 difference SE 約 $\sqrt{1/(N1-3)+1/(N2-3)}$ 。paired/dependent correlations 需要 Meng-Rosenthal-Rubin 類方法或 bootstrap。Kappa / Fleiss Kappa 則建議用 asymptotic SE 或 rater-level simulation。

4.7 Aggregated composite metrics

source-confirmed evaluate_chatgpt_meow_blind_batch.py 的 summary 會輸出 judge_average_scores；evaluate_llm.py 也會把所有 dimension scores 合併算 overall_stats。這些 aggregation 可降低單一 dimension noise，但會犧牲 interpretability，且 dimensions 相關時不能把 dimension count 當作獨立 N。若 composite 是 primary endpoint，應先在每篇 paper 內計算 composite，再對 paper-level composite 做 paired test。

4.8 Multiple comparisons and false positive risk

theoretical-assumption 若同時檢定 6 個 judge dimensions、Structural Distance、Reference Reward、success rate 等，unadjusted p-values 會提高 family-wise false positive risk。Bonferroni 使用 α/m ，Holm 是 step-down FWER 控制，Benjamini-Hochberg 控制 FDR。若只有一個 primary endpoint，其餘為 secondary/exploratory，可以把 primary 不調整，secondary 用 Holm 或 BH。

5. Sample-size and detectable-gap curves

theoretical-assumption 以下曲線都以 paper-level N 為橫軸，two-sided $\alpha=0.05$ 、power=0.8 為主。圖上垂直線標出 N=4 與 N=100。N=4 的曲線位置說明：這類 benchmark size 只能檢出非常大的差異；N=100 才開始能討論 moderate-to-small effect。

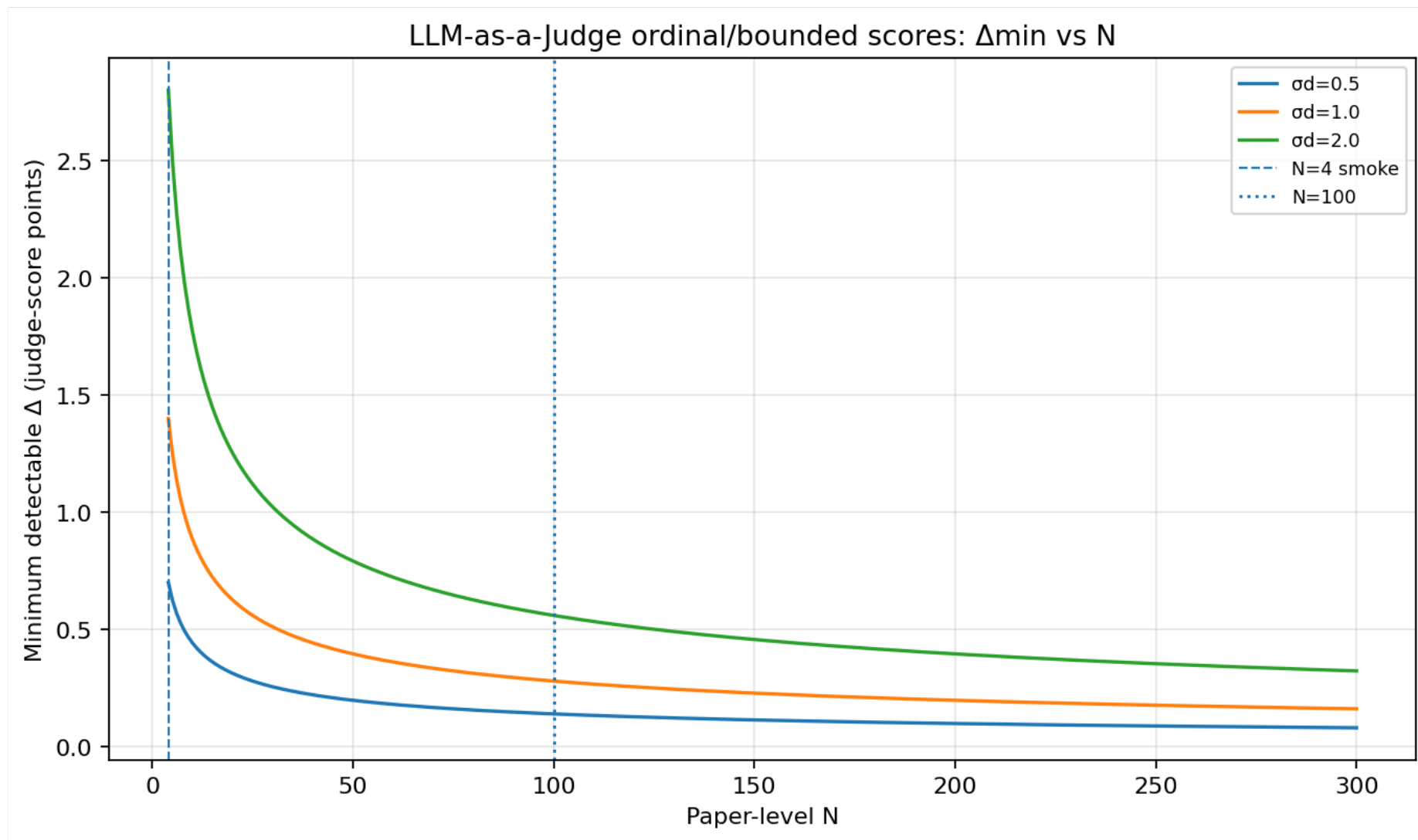


圖 1. LLM-as-a-Judge score 的 $\Delta_{\min}$ vs N

三條曲線代表 paired difference standard deviation $\sigma_d = 0.5$ 、 1.0 、 2.0 score points。若 $\sigma_d = 1.0$ ， $N=4$ 的 MDE 約 1.40 分； $N=100$ 約 0.28 分。這些分數差是可檢出差距，不代表實際效果大小。

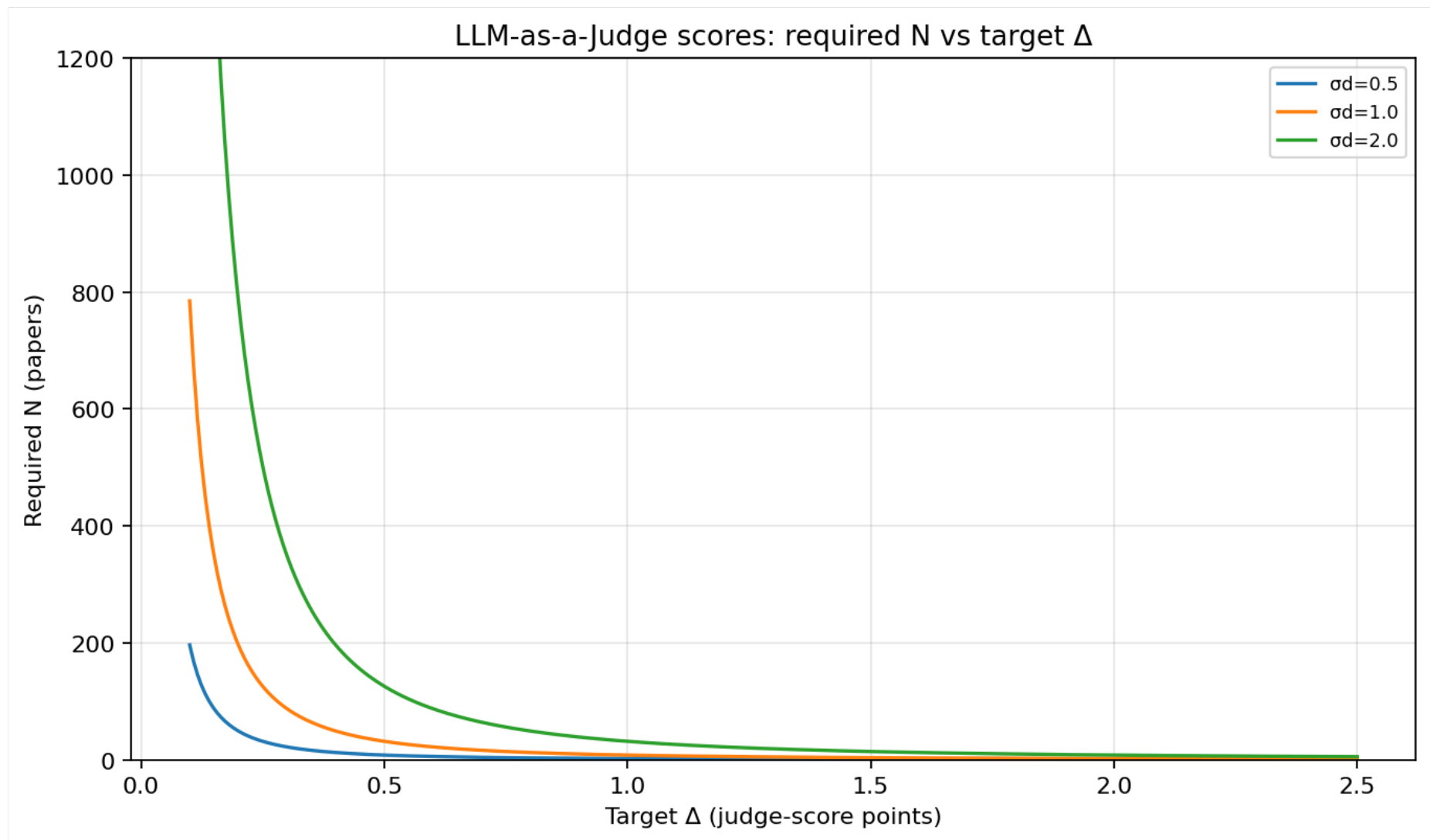


圖 2. LLM-as-a-Judge score 的 $N_{\min}$ vs target Δ

曲線從另一方向回答：若希望檢出 0.5 分差， $\sigma d=1.0$ 需要約 32 篇 papers； $\sigma d=2.0$ 需要約 126 篇。若只做 $N=4$ ，只有非常大的 judge-score gap 才可能有足夠 power。

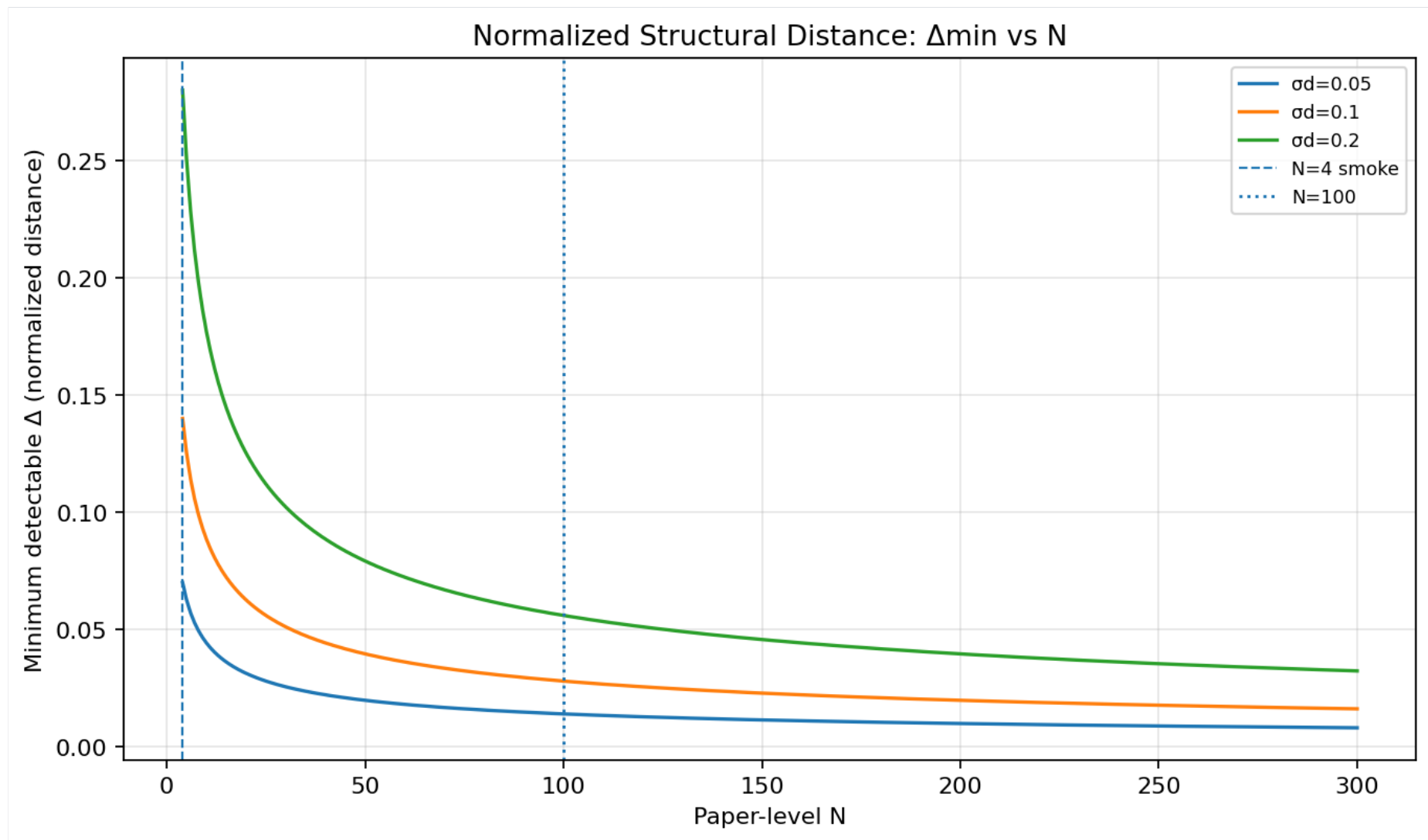


圖 3. normalized Structural Distance 的 $\Delta_{\min}$ vs N

Structural Distance 是 lower-is-better；圖中 Δ 應解讀為 distance reduction / improvement。若 $\sigma_d=0.10$ ， $N=100$ 約可檢出 0.028 normalized-distance improvement； $N=4$ 需約 0.140。

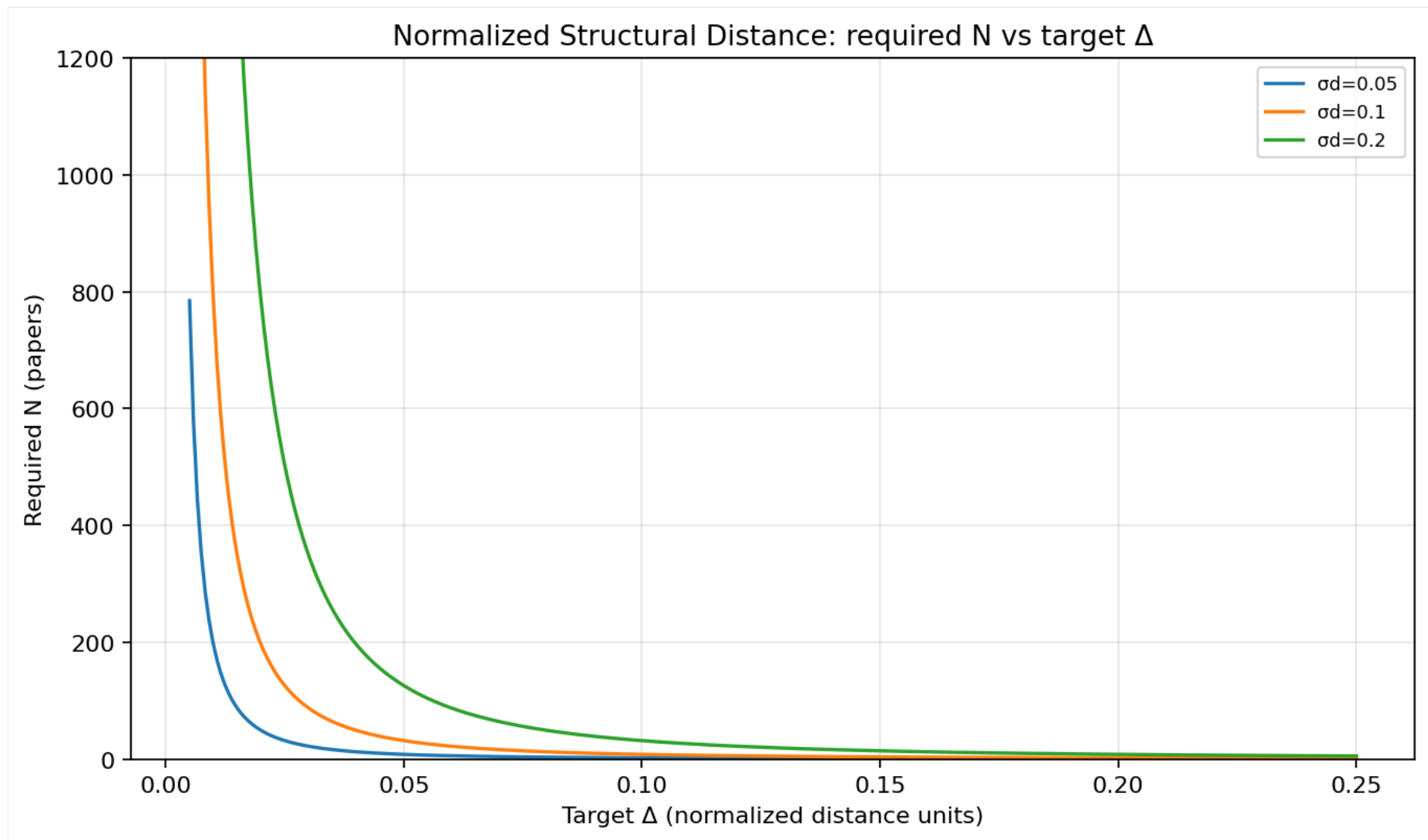


圖 4. normalized Structural Distance 的 $N_{\min}$ vs target Δ

若 target $\Delta=0.05$ 且 $\sigma_d=0.10$ ，大約需要 32 篇 papers；若 $\sigma_d=0.20$ ，需要約 126 篇。若 effect 只有 0.02，即使 $\sigma_d=0.10$ 也需約 197 篇。

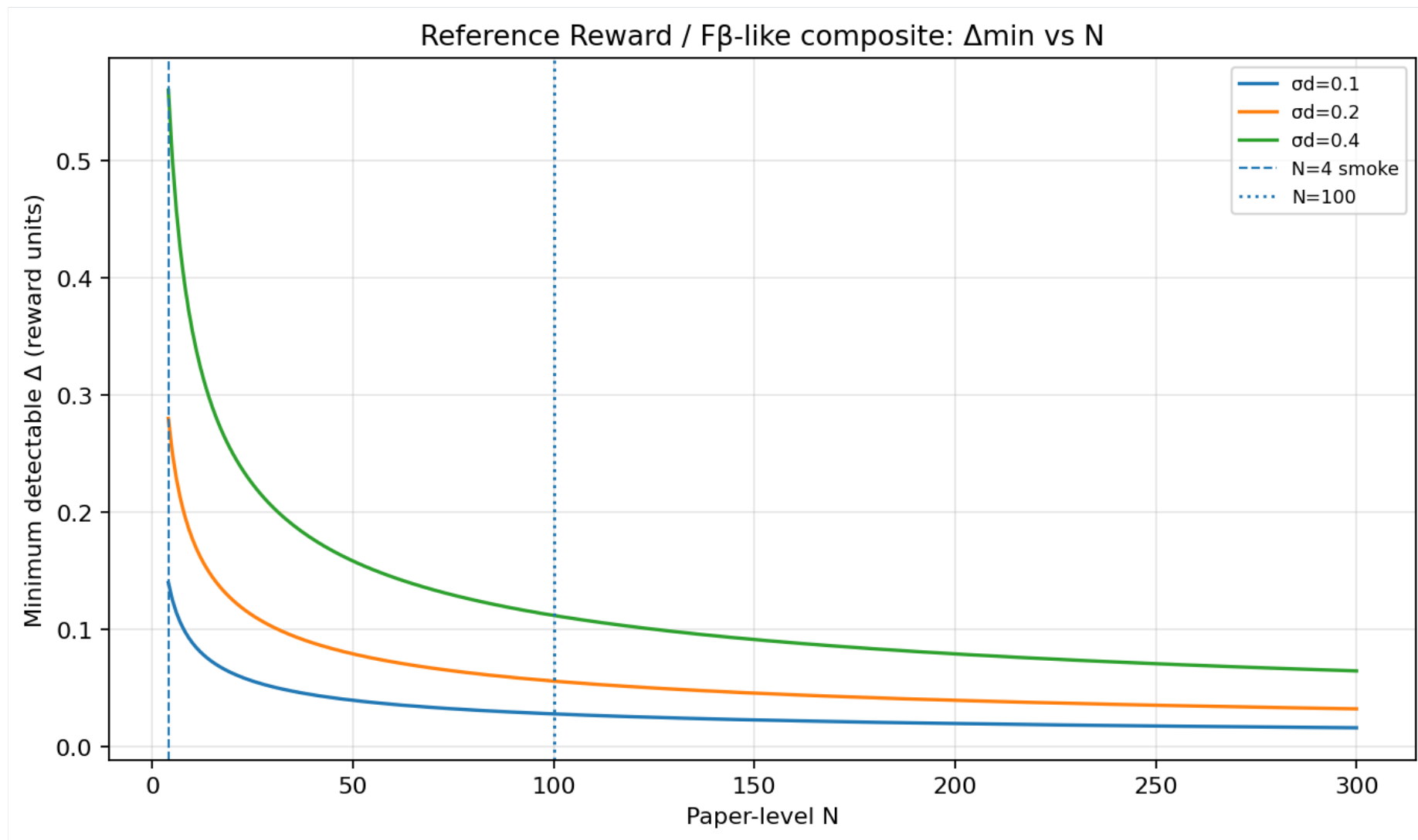


圖 5. Reference Reward / $F\beta$ -like composite 的 $\Delta_{\min}$ vs N

Reference Reward 由 set-overlap $F\beta$ 與 penalties 組成，圖中只呈現 paper-level paired difference 的連續近似。真正 planning 應搭配 confusion-count/ref-count simulation 或 hierarchical bootstrap。

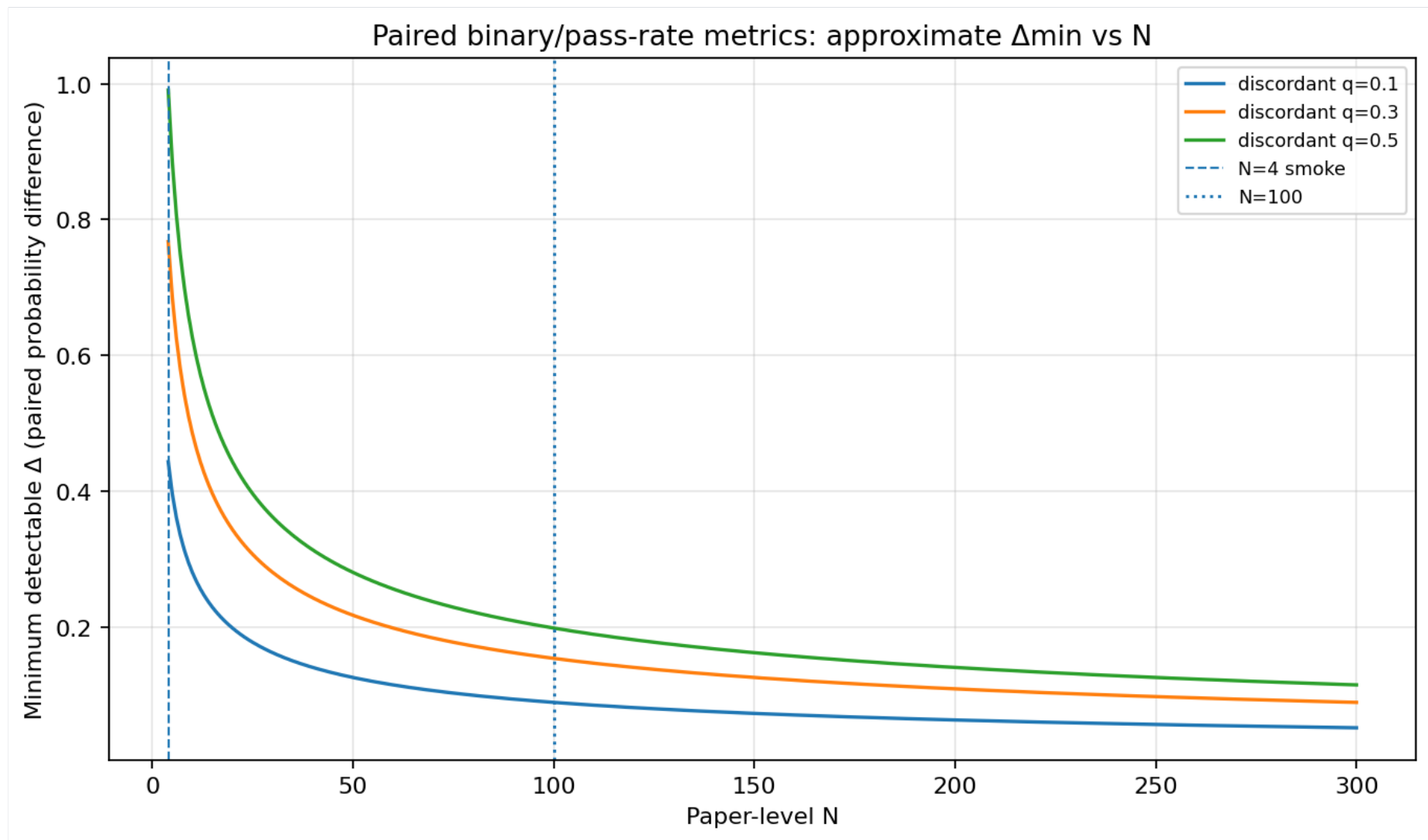


圖 6. paired binary / pass-rate-like metrics 的近似 $\Delta_{\min}$ vs N

q 是 discordant-pair fraction。若 $q=0.30$ ， $N=100$ 約可檢出 0.153 的 paired probability difference； $N=4$ 需要約 0.767，幾乎只適合 smoke check。

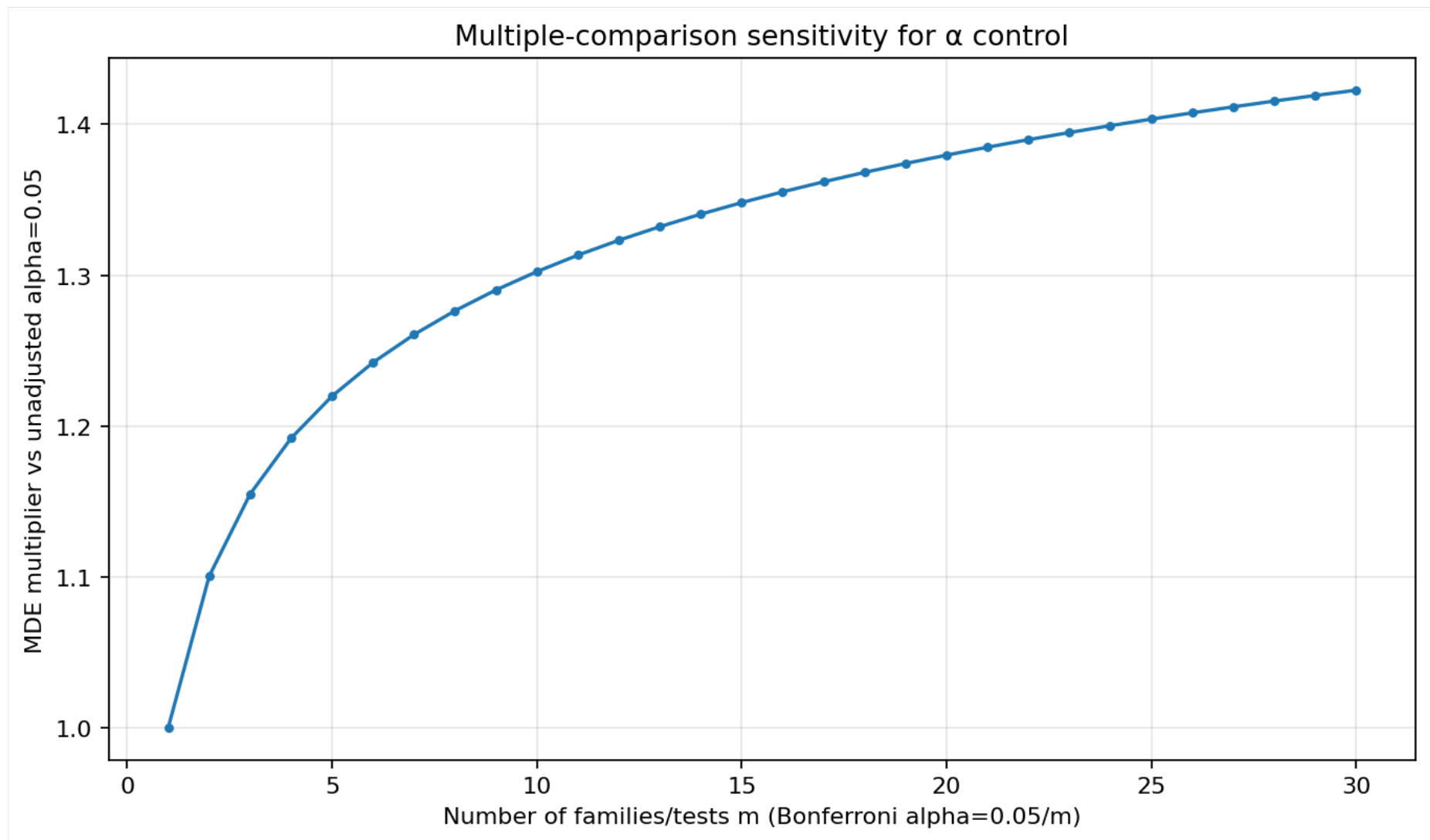


圖 7. Multiple-comparison sensitivity

Bonferroni $\alpha=0.05/m$ 會提高 MDE multiplier。 $m=6$ 時，MDE 約為 unadjusted 的 1.28 倍， N 約需 1.64 倍才能維持相同 target Δ 與 power。Holm 通常比 Bonferroni 不那麼保守，但事前 sample-size planning 可用 Bonferroni 作 worst-case。

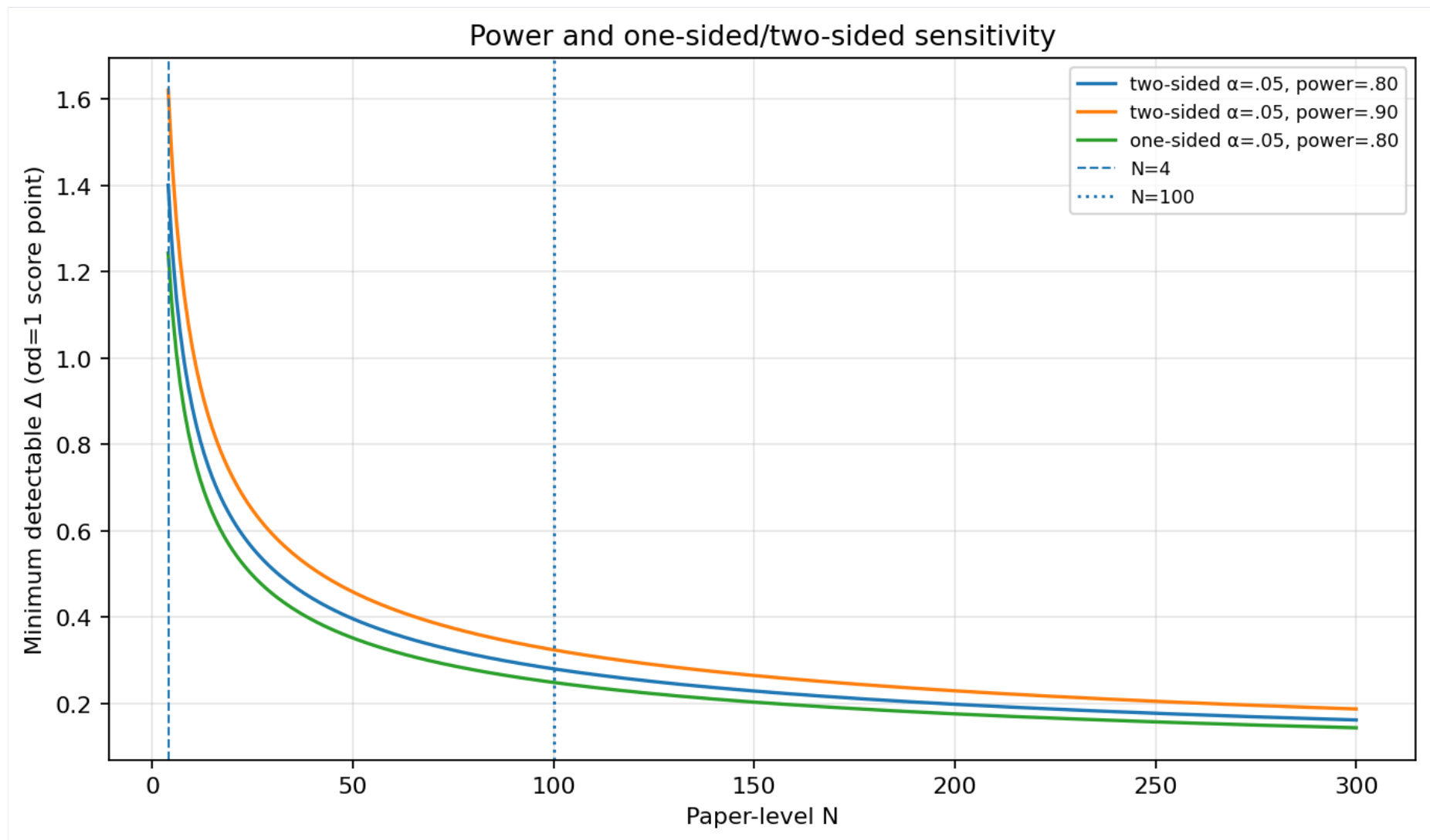


圖 8. power=0.8/0.9 與 one-sided/two-sided sensitivity

此圖固定 $\sigma d=1$ score point。power=0.9 曲線比 power=0.8 高；one-sided $\alpha=0.05$ 比 two-sided $\alpha=0.05$ 需要較小 Δ ，但只有在方向明確且 preregistered 時才應使用。

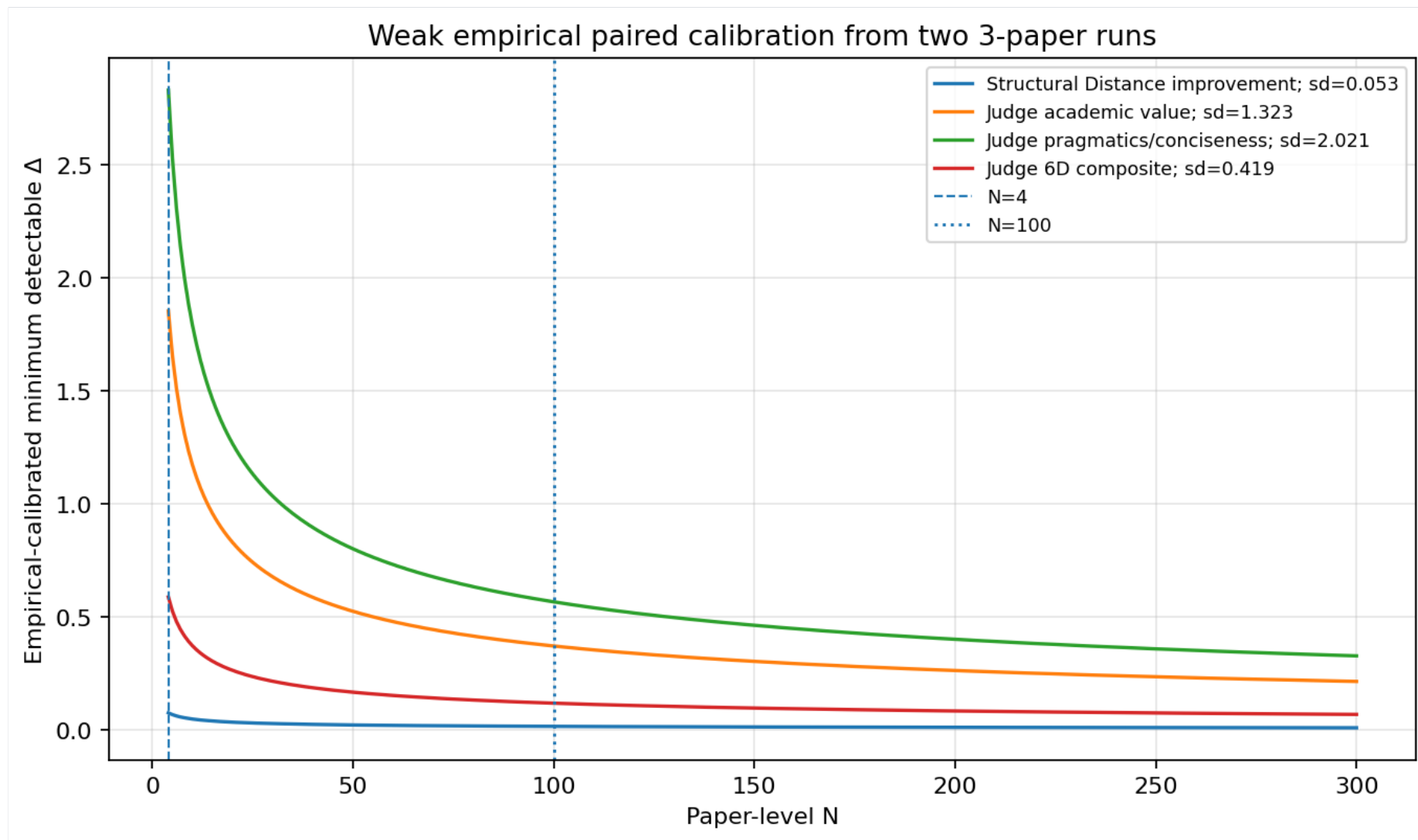


圖 9. Weak empirical paired calibration from existing runs

empirical-estimated 這張圖使用兩個 3-paper runs 的 paired od；因 $N=3$ ，估計不穩定，只能當 calibration hint。Structural Distance 與 judge-score dimensions 的 scale 不同，因此應分別解讀，不應直接比較曲線高低。

Table: N = 4, 10, 25, 50, 100, 200 時的 detectable Δ

theoretical-assumption 表 2 使用 paired continuous approximation。每格是 two-sided $\alpha=0.05$ 、power=0.8 的 $\Delta_{\min}$ 。Binary $q=0.30$ 欄用 McNemar-style approximation，單位是 paired probability difference。

N	k/sqrt(N)	Judge $\sigma d=0.5$	Judge $\sigma d=1.0$	Judge $\sigma d=2.0$	Struct $\sigma d=0.05$	Struct $\sigma d=0.10$	Struct $\sigma d=0.20$	Reward $\sigma d=0.20$	Binary $q=0.30$
4	1.401	0.700	1.401	2.802	0.070	0.140	0.280	0.280	0.767
10	0.886	0.443	0.886	1.772	0.044	0.089	0.177	0.177	0.485
25	0.560	0.280	0.560	1.121	0.028	0.056	0.112	0.112	0.307
50	0.396	0.198	0.396	0.792	0.020	0.040	0.079	0.079	0.217
100	0.280	0.140	0.280	0.560	0.014	0.028	0.056	0.056	0.153
200	0.198	0.099	0.198	0.396	0.010	0.020	0.040	0.040	0.109

Table: common target Δ 下需要的 N

theoretical-assumption 表 3 列出常用 target Δ 的 $N_{\min}$ 。N 是 paper-level paired N；若實際設計是 independent two-sample，約需把 variance term 乘上 2，或使用本報告公式中的 independent two-sample version。

metric_family	scenario	target_delta	unit	N_min_alpha05_power08	N_min_alpha05_power09
LLM judge score 0-10	medium $\sigma d=1.0$	0.250	score points	126	169
LLM judge score 0-10	medium $\sigma d=1.0$	0.500	score points	32	43
LLM judge score 0-10	medium $\sigma d=1.0$	1.000	score points	8	11
LLM judge score 0-10	medium $\sigma d=1.0$	1.500	score points	4	5
LLM judge score 0-10	medium $\sigma d=1.0$	2.000	score points	2	3
LLM judge score 0-10	high $\sigma d=2.0$	0.250	score points	503	673
LLM judge score 0-10	high $\sigma d=2.0$	0.500	score points	126	169
LLM judge score 0-10	high $\sigma d=2.0$	1.000	score points	32	43
LLM judge score 0-10	high $\sigma d=2.0$	1.500	score points	14	19
LLM judge score 0-10	high $\sigma d=2.0$	2.000	score points	8	11

metric_family	scenario	target_delta	unit	N_min_alpha05_power08	N_min_alpha05_power09
Normalized Structural Distance	medium $\sigma d=0.10$	0.010	distance units	785	1051
Normalized Structural Distance	medium $\sigma d=0.10$	0.020	distance units	197	263
Normalized Structural Distance	medium $\sigma d=0.10$	0.050	distance units	32	43
Normalized Structural Distance	medium $\sigma d=0.10$	0.100	distance units	8	11
Normalized Structural Distance	medium $\sigma d=0.10$	0.200	distance units	2	3
Normalized Structural Distance	high $\sigma d=0.20$	0.010	distance units	3140	4203
Normalized Structural Distance	high $\sigma d=0.20$	0.020	distance units	785	1051
Normalized Structural Distance	high $\sigma d=0.20$	0.050	distance units	126	169
Normalized Structural Distance	high $\sigma d=0.20$	0.100	distance units	32	43
Normalized Structural Distance	high $\sigma d=0.20$	0.200	distance units	8	11
Reference Reward / bounded ratio	medium $\sigma d=0.20$	0.050	reward units	126	169
Reference Reward / bounded ratio	medium $\sigma d=0.20$	0.100	reward units	32	43
Reference Reward / bounded ratio	medium $\sigma d=0.20$	0.200	reward units	8	11
Reference Reward / bounded ratio	medium $\sigma d=0.20$	0.300	reward units	4	5
	medium $\sigma d=0.20$	0.500	reward units	2	2

metric_family	scenario	target_delta	unit	N_min_alpha05_power08	N_min_alpha05_power09
Reference Reward / bounded ratio					
Reference Reward / bounded ratio	high $\sigma d=0.40$	0.050	reward units	503	673
Reference Reward / bounded ratio	high $\sigma d=0.40$	0.100	reward units	126	169
Reference Reward / bounded ratio	high $\sigma d=0.40$	0.200	reward units	32	43
Reference Reward / bounded ratio	high $\sigma d=0.40$	0.300	reward units	14	19
Reference Reward / bounded ratio	high $\sigma d=0.40$	0.500	reward units	6	7
Paired binary / pass-rate-like outcome	discordant fraction $q=0.30$	0.020	probability difference	5887	7881
Paired binary / pass-rate-like outcome	discordant fraction $q=0.30$	0.050	probability difference	942	1261
Paired binary / pass-rate-like outcome	discordant fraction $q=0.30$	0.100	probability difference	236	316
Paired binary / pass-rate-like outcome	discordant fraction $q=0.30$	0.150	probability difference	105	141
Paired binary / pass-rate-like outcome	discordant fraction $q=0.30$	0.200	probability difference	59	79

Weak empirical variance-calibrated table

empirical-estimated 表 4 來自兩個 representative results summary JSON 的 paired differences：gpt-5.4-xhigh-3papers vs gemini-3.1-pro-preview-3papers。正值代表前者較好；Structural Distance 已轉成 lower-is-better improvement。N=3 太小，因此這個表只用於說明如何從 existing runs 估 σd 。

metric	n_pairs	mean_diff	sd_paired_diff	mde_N4_alpha05_power08	mde_N25_alpha05_power08	mde_N100_alpha05_power08
	3	0.106	0.053	0.075	0.030	0.015

metric	n_pairs	mean_diff	sd_paired_diff	mde_N4_alpha05_power08	mde_N25_alpha05_power08	mde_N100_alpha05_power08
Structural Distance improvement (B - A; lower-is-better converted)						
LLM judge dimension: 结构_信息快速定位	3	0.000	1.000	1.401	0.560	0.280
LLM judge dimension: 结构_详略得当	3	0.000	1.000	1.401	0.560	0.280
LLM judge dimension: 内容_章节互斥性	3	0.333	0.577	0.809	0.323	0.162
LLM judge dimension: 内容_逻辑深度	3	0.000	1.000	1.401	0.560	0.280
LLM judge dimension: 内容_学术价值	3	1.500	1.323	1.853	0.741	0.371
LLM judge dimension: 语用_描述性与简洁性	3	-1.167	2.021	2.831	1.132	0.566
LLM judge composite mean of 6 dimensions	3	0.111	0.419	0.588	0.235	0.118

6. Practical interpretation for Outline_COT / MEOW evaluation

6.1 N=4 能支持什麼結論？

theoretical-assumption N=4 的 paired continuous MDE 是 $1.401 \sigma_d$ 。若 judge-score paired $\sigma_d=1.0$ ，必須有約 1.4 分以上的差距才可能達到 power=0.8；若 $\sigma_d=2.0$ ，需約 2.8 分。這代表 N=4 可以用來發現 catastrophic regression、smoke failure、或非常大的 qualitative difference，但不能用來宣稱一般大小的 model/prompt improvement。

6.2 N=10、25、50 的定位

theoretical-assumption N=10 仍偏 exploratory，MDE 約 $0.886 \sigma_d$ 。N=25 時，MDE 約 $0.560 \sigma_d$ ，開始能看到 medium effect。N=50 時，MDE 約 $0.396 \sigma_d$ ，對 judge $\sigma_d=1.0$ 的 metric 約可檢出 0.4 分；對 normalized distance $\sigma_d=0.10$ 約可檢出 0.04。

6.3 N=100 與 N=200 的定位

theoretical-assumption N=100 的 MDE 約 $0.280 \sigma_d$ ；N=200 約 $0.198 \sigma_d$ 。若要支撐跨模型、跨 prompt、或多 metric 的穩健 claim，N=100 是合理的最低嚴肅 benchmark scale；N=200 更適合小效果、multiple comparisons、或需要 subgroup/facet analysis 的 setting。

6.4 對 lower-is-better metric 的報告方式

source-confirmed Structural Distance 是 lower-is-better。報告時建議把差異寫成 distance reduction，例如 $\Delta = \text{mean}(\text{distance_baseline} - \text{distance_new})$ 。這樣正值代表新方法變好，與 judge-score higher-is-better 的符號一致，避免讀者誤解。

6.5 Gold outline mismatch 與 construct validity

theoretical-assumption Structural Distance 只度量 tree shape；reference reward 偏向 numbering/ref overlap；LLM judge 則偏向標題語義與寫作品質。三者測到的 construct 不同。若某模型 Structural Distance 變好但 judge score 變差，可能代表它模仿 reference shape 卻犧牲內容品質；若 judge score 變好但 distance 變差，可能代表它寫了合理但不同形狀的 outline。這不是單純統計問題，而是 metric construct validity 問題。

7. Recommendations

theoretical-assumption 一、把 primary endpoint 限定為一個。若目標是 faithful reconstruction，可選 normalized Structural Distance；若目標是 reader-facing outline quality，可選 6D judge composite mean 或一個 preregistered primary dimension。其餘 metrics 應作為 secondary / diagnostic outputs。

source-confirmed 二、不要把 upstream 6D judge 與 paper 5D judge 混稱。報告中應明確寫「upstream repo-defined 6D judge」，並標出新增的內容_学术价值 dimension。若要對齊 paper main table，必須補 source-verified 5D prompt 或重新實作 5D metric。

theoretical-assumption 三、比較兩個模型 / prompt / run 時，建議最少 N=25 作 pilot；N=50 作中等效果 claim；N=100 作主要 benchmark claim；N=200 作小效果、多模型、多 metric 或 subgroup analysis。N=4 僅可稱 smoke / exploratory。

theoretical-assumption 四、每個 metric 都應報告 paired mean difference、95% CI、effect size Δ/σ_d 、p-value、N、missing/failure counts。若用 LLM judge，應報告 judge model、prompt version、parse failure、是否 repeated judging、temperature / deterministic settings。

theoretical-assumption 五、Reference Reward / F β -like metrics 應用 hierarchical bootstrap 或 simulation over ref-overlap scenarios。不要把 section-level F β 當獨立 observations；paper-level article_reward 才是主分析單位。

theoretical-assumption 六、若同時看 6D judge dimensions，建議先 report composite primary，再對 dimensions 做 Holm adjusted secondary analysis；若研究目標是探索哪個 dimension 改善，可用 BH FDR 控制並把結果標為 exploratory。

8. Limitations

theoretical-assumption 一、LLM judge noise：single-call judge score 包含模型評審的隨機性、prompt sensitivity、API backend behavior。若沒有 repeated judge calls， σ_d 會混合 paper variability 與 judge noise。

theoretical-assumption 二、paired dependence：同一 paper 的兩個 outputs 共享 topic difficulty、reference outline、paper length 和 domain；這正是 paired design 的優點，但分析時也必須保留 pairing。

theoretical-assumption 三、multiple comparisons：多模型、多 prompt、多 run、多 dimensions 一起比較時，如果只報最顯著的結果，false positive risk 很高。應事前指定 primary endpoint 和 correction plan。

theoretical-assumption 四、gold outline mismatch：MEOW/reference outline 本身可能不是唯一正確大綱。Structural Distance 對 alternative but valid structures 可能過度懲罰；LLM judge 則可能 reward plausible but non-faithful outlines。

theoretical-assumption 五、metric construct validity：tree shape、ref overlap、reader-facing quality、academic value、pragmatic concision 是不同 construct。單一 aggregate 不應取代 error analysis。

empirical-estimated 六、existing-run empirical variance 目前只從 $N=3$ paired results 做示範估計，精度很差。正式 planning 應用更大的 pilot 或歷史 paired runs 重新估計 σ_d 。

9. Appendix

9.1 Assumption table

assumption	value	status
Primary design	paired comparison on the same paper set; N is number of papers	theoretical-assumption aligned with repo output schema
Main alpha/power	two-sided $\alpha=0.05$, power=0.8; sensitivity at power=0.9 and one-sided $\alpha=0.05$	theoretical-assumption
Continuous approximation	paired differences approximately normal for planning; small- N inference should use permutation/bootstrap	theoretical-assumption

assumption	value	status
LLM judge scale	0-10 half-point ordinal scores are treated as interval only for primary planning; ordinal/robust checks required	source-confirmed + theoretical-assumption
Cluster unit	sections, nodes, and reference pairs are nested within papers and are not independent N	theoretical-assumption / experimental-design requirement
Empirical curves	two existing 3-paper runs provide paired σ_d estimates; these are weak and not stable enough for final power planning alone	empirical-estimated

9.2 Formula notes

theoretical-assumption Paired t-test 的 closed-form planning 公式是假設 d_i 的 sampling distribution 近似 normal。當 N 小、分布有長尾、或 metric 有邊界堆積時，公式只作 rough planning。正式分析可用 paired permutation test 直接在 observed d_i 上隨機翻號；CI 可用 paired bootstrap over papers。

theoretical-assumption For ratio metrics such as $F\beta$, a delta-method approximation can be constructed by writing $F\beta$ as a differentiable function of TP, FP, FN and applying $\text{Var}(g(X)) \approx \nabla g^T \Sigma \nabla g$. In practice, because section/ref counts are nested and sparse, hierarchical bootstrap or simulation is usually safer.

9.3 Python code excerpt

完整 Python build script 已放在附件 zip；下列是核心 sample-size functions。

```

from statistics import NormalDist
import math

z = NormalDist().inv_cdf

def k_factor(alpha=0.05, power=0.8, two_sided=True):
    z_alpha = z(1 - alpha/2) if two_sided else z(1 - alpha)
    return z_alpha + z(power)

def paired_continuous_delta_min(N, sigma_d, alpha=0.05, power=0.8):
    return k_factor(alpha, power, True) * sigma_d / math.sqrt(N)

def paired_continuous_N_min(delta, sigma_d, alpha=0.05, power=0.8):
    return (k_factor(alpha, power, True) * sigma_d / delta) ** 2

def independent_equal_n_delta_min(n_per_arm, sigma, alpha=0.05, power=0.8):
    return k_factor(alpha, power, True) * math.sqrt(2) * sigma / math.sqrt(n_per_arm)

def paired_binary_delta_min(N, q_discordant, alpha=0.05, power=0.8):
    # Approximate planning formula for McNemar-style paired pass/fail outcome.
    return k_factor(alpha, power, True) * math.sqrt(q_discordant) / math.sqrt(N)

```

9.4 Generated artifacts

本次產出的附件包含：metric_inventory.csv、detectable_delta_by_N.csv、required_N_by_target_delta.csv、empirical_weak_paired_sd_estimates.csv、source_provenance.csv、assumption_table.csv、所有 PNG 圖，以及完整 build_metric_power_report.py。